Mathematics – Linear Algebra

1 Prerequisites to Linear Algebra

1.1 Coordinate Systems in the 3-Space (3D Space)
1.2 Slope-Intercept and Normal Equations of Lines in 2-Space (2D Plane)
1.3 Vectors and Scalars
1.4 Adding and Scaling Vectors
1.5 Linear Combinations
1.6 Matrices
1.7 Linear Transformations and Maps
1.8 Matrix-Vector Multiplication
1.9 Refresher: Computation Laws of Reals
1.10 Pythagorean Theorem and Distance between Points
1.11 Pythagorean Identity

2 Systems of Linear Equation and Geometric Interpretation

2.1 Ways of Looking at Equations
2.2 Linear and Non-Linear Solutions
2.3 Systems of Linear Equations
2.4 Solution Sets
2.5 2x2 Systems
2.6 3x2 Systems
2.7 3x3 Systems
2.8 2x3 Systems
2.9 General Case: m x n Systems

3 Solving Systems of Linear Equations

3.1 Systems of Linear Equations: Algebraic Solution
3.2 Elementary Row Operations
3.3 Gaussian Elimination vs. Gauss-Jordan Elimination
3.4 Elimination: Matrix Form / Coefficient and Augmented Matrices
3.5 Constructing Augmented Matrices from Systems of Linear Equations
3.6 Consistency of Systems
3.7 General Reduction Algorithm
• REF vs. RREF Forms